

# **NFC Card and Tag Security Enhancement Framework**

## **Using Hybrid Detection**

**NFC Tag Data Integrity Verification**

J26-CS-316

**Project Proposal Report**

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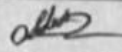
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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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## ABSTRACT

Access control, identification, asset tracking, product authentication, and information-sharing systems all make extensive use of Near **Field Communication (NFC)** cards and tags. Nonetheless, a lot of NFC cards and tags have writable memory, which enables the modification of application data, raw memory, or security-related data without affecting the card's ability to function normally. While a straightforward whole-memory hash comparison can simply indicate whether the current state is the same or different from the original state, conventional NFC scanners mostly display the current stored data. It is unable to pinpoint exactly what was altered, where it happened, or which area of the card was impacted. This paper suggests a multi-layer integrity-verification technique for NTAG215, MIFARE Classic 1K, and MIFARE DESFire test cards: NFC Tag Data Integrity Verification with Tamper Localization. The initial card state, which includes application/NDEF data, raw-memory data, and supported security-state information, is saved as a trusted baseline and secured with HMAC upon authorized enrollment. Three layers—application/NDEF verification, raw-memory verification, and security-state verification—are used to compare the present card state with this baseline during subsequent verification. When a discrepancy is found, the system locates the field, page, block, file, or logical memory region that is impacted and, if available, records the previous and current values. In order to detect authorized updates, unjournalled modifications, rollback situations, and inconsistent journal records, the system also makes use of a trusted backend integrity journal and supports compact card-resident evidence. Localization accuracy, tamper-class accuracy, false-modification rate, journal verification accuracy, cross-reader consistency, UNKNOWN rate, and verification latency will all be used to assess the suggested approach in a controlled laboratory setting.

Keywords— NFC, Data Integrity, Tamper Detection, Tamper Localization, Integrity Verification.

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## LIST OF ABBREVIATIONS

| Abbreviation | Meaning                                  |
|--------------|------------------------------------------|
| AES          | Advanced Encryption Standard             |
| AID          | Application Identifier                   |
| API          | Application Programming Interface        |
| CMAC         | Cipher based Message Authentication Code |
| CSV          | Comma Separated Values                   |
| HMAC         | Hash based Message Authentication Code   |
| MAC          | Message Authentication Code              |
| NDEF         | NFC Data Exchange Format                 |
| NFC          | Near Field Communication                 |
| NVM          | Non Volatile Memory                      |
| OTP          | One Time Programmable                    |
| RFID         | Radio Frequency Identification           |
| SHA-256      | Secure Hash Algorithm 256-bit            |
| TLV          | Type Length Value                        |
| UID          | Unique Identifier                        |

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# 1 INTRODUCTION

## 1.1 background and Context

Access control, identification, transportation, ticketing, product authentication, asset management, smart labeling, and digital information sharing systems all make use of Near Field Communication (NFC), a significant short range communication technology. Numerous types of data, such as identifiers, NDEF formatted records, application data, raw memory values, configuration parameters, access control information, and security related states, can be stored on NFC enabled cards and tags. The integrity of the stored data is still a major security risk, despite the fact that these technologies enable easy and affordable interaction between a card or tag and a scanner. Certain memory regions might still be editable after the card has been issued, depending on the card family and its setup. Because of this, even after some of its recorded data has been altered, an NFC card or tag can still react to a reader in a typical manner. As a result, a successful card read does not guaranty that the trusted information that was initially registered is still the same as the data that is currently stored on the card. This issue is especially significant in instances where businesses use the data on the card or tag to make decisions about operations, identification, or security.

This issue affects a number of sectors and user groups. Access control operators and security administrators need to be certain that a given credential still contains the original, reliable data. NFC tags are used by asset and inventory managers to link physical objects to backend information. Even if the tag is still readable, unauthorized modifications to identifiers or application data may result in inaccurate asset records. NFC tags can be used by product-authentication and brand protection systems to hold product identifiers, web URLs, or verification information; if these values are changed, consumers could be sent to unapproved material or obtain inaccurate product information. A simple signal that the card is "different" following a suspected modification is inadequate evidence to identify the type and location of the alteration, which also affects security auditors and digital forensic investigators. Recent study reflects the broader significance of RFID/NFC security. According to a systematic review of RFID based access control security, MIFARE technologies are widely used worldwide, with over 1.2 billion users spread across about 750 cities. Attacks like cloning, skimming, replay, cryptanalysis, brute force attacks, and other cyber physical threats are identified as ongoing security concerns. Empirical research on MIFARE Classic has also demonstrated that stored card information can be selectively modified in operational environments when legacy protection mechanisms, weak keys, or incorrect access settings are present, and the study recommends measures such as application layer integrity validation and backend binding to strengthen trust in stored credential data.

The project evidence that is currently available for Sri Lanka does not offer a confirmed national number that describes the percentage of deployed tags that are still readable or the frequency of NFC tag tampering. In order to give local contextual evidence, this study will refrain from making unsupported local prevalence claims and instead depend on controlled experimental evidence and, when ethically permitted, limited read only observation of authorized NFC deployments. While certain aspects of this issue are addressed by current security techniques, they fall short of providing the kind of integrity evidence that this study demands. Although NFC memory can be permanently locked to prevent future changes, this

method is inappropriate when valid updates are needed and does not assist tags that were previously deployed in a writable form. Manufacturer originality signatures do not demonstrate that user or application data has not changed after issue, but they can offer proof on the provenance of enabled chips Password, key, or access right based protection can limit who is permitted to execute write operations, but if access protection is circumvented or credentials that are properly held are exploited, these techniques may not always give an investigator comprehensive proof of what was altered. Although many of these methods focus on the visible application or NDEF content and may not examine unused raw memory or security-related configuration sections, backend content comparison can save an expected value and compare it with subsequent reads. Change detection is enhanced by a straightforward whole memory SHA-256 comparison as any altered byte results in a new digest, but the output is still merely a match or mismatch. It is unable to pinpoint the precise page, block, file, field, or logical area that has been altered. The integrity component was initially defined as a SHA-256 comparison of authorized NDEF and memory snapshots that produced VALID, MODIFIED, or UNKNOWN; the proposed research extends this fundamental concept into a more thorough region-aware verification and tamper-localization method.

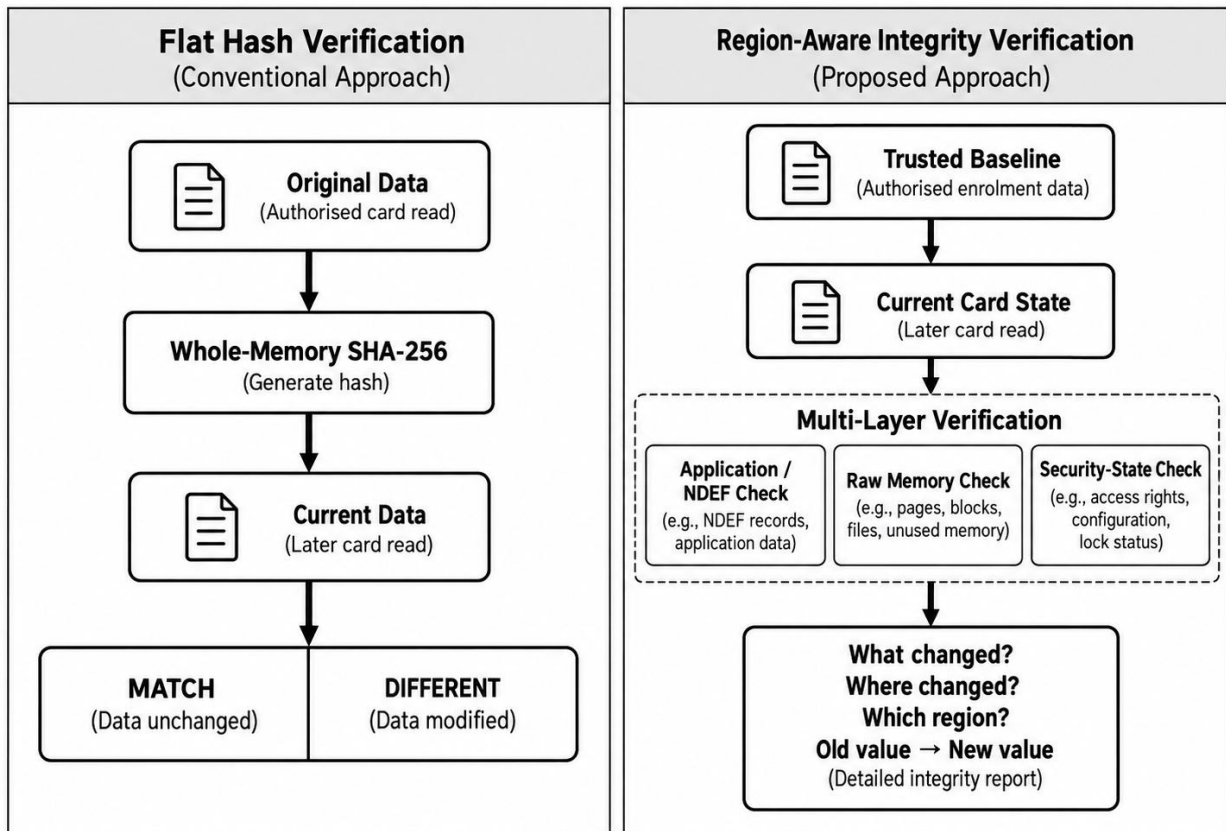


Figure 1 Comparison of Flat Hash Verification and Region-Aware Integrity Verification

In order to get beyond these restrictions, the proposed study, NFC Tag Data Integrity Verification with Tamper Localization, compares the present data state of authorized NFC cards to a safe, reliable baseline and provides comprehensive proof when a change is detected. The method is designed to handle several NFC card families, such as MIFARE Classic 1K, MIFARE DESFire, and NTAG215, while honoring the distinct memory architectures of each family. The system reads the original card, parses and canonicalizes the available data, determines the card family, organizes the data into logical areas, and establishes a reliable baseline for the first authorized enrollment. Evidence about application or NDEF data, raw memory, and supported security state data are all included in the baseline. HMAC is used to secure the stored baseline so that any unauthorized changes to the trusted reference can be identified prior to verification. Three primary verification layers application/NDEF verification, raw memory verification, and security state verification are used to compare the present card state with the trusted baseline for subsequent verification. When a difference is found, the system makes an effort to determine what changed, where it happened, which logical region was impacted, and, if available, the previous and current values. Localization can be stated in pages for NTAG215, sectors and blocks for MIFARE Classic 1K, and applications, files, records, or offsets that are legally readable for MIFARE DESFire. Compared to a single whole memory hash mismatch, this family aware approach enables the verification method to generate more valuable evidence.

Additionally, tamper evident integrity event recording is introduced by the suggested method. The entire trusted history of enrollment, verification, and authorized update events will be kept in an independent backend journal, and where the card family, memory, and security measures allow it, compact card resident integrity information will be kept. Because of this design, conditions like an authorized update, a modification that does not match a reliable journal event, missing journal information, an inconsistent journal state, or a rollback of previously recorded integrity evidence can all be detected by the verification process. Additionally, the system will differentiate between the initial detection of an unknown external modification and the precise moment of a trusted system controlled update. The research will not promise to recover the precise time of an unseen external alteration because passive NFC cards do not independently maintain a reliable real time clock. Rather, the system will provide a defensible possible modification window by recording the last known valid time and the first detected modified time. Along with supporting details including the impacted area, page/block/file location, past and present values, journal status, reader information, event sequence, and pertinent timestamps, the verification findings will be given as VALID, MODIFIED, or UNKNOWN. Determining if a byte identical physical copy is the original physical card stays outside of this component and belongs to the separate clone/provenance analysis within the entire project architecture. This keeps the component primarily focused on data state integrity.

Owned or specially authorized NFC test cards and readers will be used in a controlled laboratory approach to assess the proposed work. In order to assess whether the system accurately localizes and classifies each change, known adjustments will be purposefully inserted so that the precise ground truth is available. Because an integrity system must show that routine readings are not mistakenly reported as modifications, unchanged-card situations will also be provided. Therefore, in addition to whether changes are detected, the suggested method will be assessed for tamper localization accuracy, modification class accuracy, false

modification rate, behavior under incomplete or unreliable reads, journal-verification accuracy, cross reader consistency, UNKNOWN rate, and verification latency. Compared to only showing that a cryptographic hash changes when the input changes, this offers a more substantial scientific contribution. The entire study falls under the cybersecurity category and adds to the broader NFC Card and Tag Security Enhancement Framework Using Hybrid Detection, which keeps the data integrity component separate from the distinct security scoring, clone-detection, and behavioral-analysis components.

The approved project material already situates the entire investigation within the field of cybersecurity research and connects it to NFC security, smart infrastructure, and access control applications.

Additionally, the study advances pertinent Sustainable Development Goals of the United Nations. By enhancing the robustness and reliability of NFC enabled digital infrastructure utilized in identification, logistics, authentication, and smart systems, it supports SDG 9 Industry, Innovation, and Infrastructure. By enhancing the quality and auditability of identity and access related data utilized by organizations and institutions, it also contributes to SDG 16 Peace, Justice, and Strong Institutions. By increasing confidence in product and asset information, the effort can also indirectly support SDG 12 Responsible Consumption and Production where NFC tags are utilized for supply chain tracking and product verification. SDGs 9 and 16 are already identified as pertinent to the overall NFC security framework in the project's topic literature.

## **1.2 Literature Review**

Successful contactless card reading does not always ensure that the stored data is safe or unaltered, according to current studies on RFID and NFC security. Recurrent RFID/NFC risks, such as cloning, skimming, replay, cryptanalysis, brute force assaults, and data manipulation, were revealed in a recent comprehensive literature review that covered 265 research [1]. The review also highlighted the extensive use of MIFARE technology in access control environments. However, rather than offering a way to compare a card's present memory state with a trusted enrolled state and pinpoint the precise position of a modification, these research mostly categorize vulnerabilities and attack types.

Card data can be selectively altered in operational systems, according to recent empirical research on MIFARE Classic. Damasceno et al. showed that cloning and targeted change of stored card data can be made possible by flaws related to CRYPTO-1, default or weak sector keys, and improper access bit setups [2]. Additionally, their study suggests actions like backend binding of trusted credential information and application layer integrity checks. Nevertheless, rather than creating a cross family approach that localizes a detected change to a particular field, page, block, file, or logical region, the study mostly concentrates on illustrating vulnerabilities and mitigation techniques [2].

Previous NFC research has also looked into NDEF level integrity protection. To accommodate a greater variety of cryptographic techniques, such as HMAC based authentication, for safeguarding the integrity of NDEF messages, Hameed, Jamali, and Samad suggested changes to the NFC

Signature Record Type Definition [3]. According to their analysis, HMAC based methods offer relatively low processing delay and storage overhead, which makes them appropriate for limited NFC tags [3]. While Roland, Langer, and Scharinger discovered flaws including record-composition attacks in the NDEF signature system [5], related work by Roland and Langer examined digital signature records for NDEF messages [4]. Although these studies offer useful protection for NDEF information, they continue to prioritize message integrity and authentication at the application layer over complete card-state verification across raw memory and security related areas.

The significance of protecting trusted application data is further illustrated by research on real world NFC applications. A smartphone enabled smart card attendance system that combines contactless cards, readers, mobile devices, and backend services was created by Zlatarov and Ivanova [6]. According to their research, NFC card data is typically processed as a component of a larger application design rather than separately. Similar to this, Bhanushali et al.'s NFC enabled smart visiting card system shows how NFC can be used to exchange contact information, social media links, payment details, and other application data [7]. Although post-enrollment tamper localization is not directly the focus of these experiments, the accuracy of the data retrieved from the card or tag determines the system's utility in both scenarios.

A similar but distinct security goal is addressed by authentication techniques. An RFID/NFC gadget with integrated fingerprint authentication that manages card data access through biometric verification and secure internal storage was proposed by Ranasinghe and Yu [8]. These methods can limit who can access or use a device, but successful authentication does not guarantee that the raw memory, stored application data, or security configuration has not changed since enrollment. Consequently, rather than being comparable procedures, integrity verification and authentication should be viewed as complimentary security roles [8].

Because a digest can be computed from a trusted state and compared with the digest of a later state, cryptographic hashing is generally useful to integrity checking. The NIST Secure Hash Standard [12] defines SHA-256, whereas HMAC offers keyed message authentication to safeguard stored integrity evidence [13]. Nevertheless, a single whole state SHA-256 hash simply yielded an equality or inequality result; the digest does not identify the specific area of the NFC memory that was mismatched. While direct page, block, or file comparison is still a crucial benchmark for small NFC memory, hierarchical hashing methods based on Merkle's work can organize integrity evidence into smaller sections [11].

Integrity verification method design is also impacted by card family variances. Page oriented memory is used by NTAG215 [14], sector and block oriented memory with sector level access requirements is used by MIFARE Classic 1K [15], and an application- and file-oriented architecture with more robust access control mechanisms is used by MIFARE DESFire [16]. As a result, without family specific interpretation, a verification technique created for a single fixed memory architecture cannot be easily applied to all three technologies. As a result, the suggested study maintains the native memory structure of each card family while employing a common logical verification procedure.

Vulnerability detection, authentication, NDEF message protection, digital signatures, and cryptographic integrity checks have all been covered in NFC security research, according to the literature [1]-[8]. These methods, however, only partially support a unified mechanism that looks

at application/NDEF data, raw memory, and supported security-state information independently, compares a later NFC card state against a protected trusted baseline, and pinpoints the precise location of a detected modification. The suggested NFC Tag Data Integrity Verification with Tamper Localization method is motivated by this constraint.

| <b>Existing Approach / Study</b>           | <b>Main Focus / Method</b>                                                        | <b>Strength</b>                                                                                        | <b>Limitation Relevant to This Research</b>                                                           | <b>Influence on Proposed Method</b>                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| RFID/NFC Security Systematic Review        | Reviews vulnerabilities and attacks in RFID/NFC access-control systems            | Provides broad evidence of cloning, replay, tampering, cryptanalysis, and other cyber-physical threats | Does not provide a detailed method to localize an exact memory modification                           | Supports the need for a defensive integrity-verification and tamper-localization mechanism |
| MIFARE Classic Tampering Research          | Empirical testing of cloning and selective card-data modification                 | Demonstrates that stored card data can be modified in practical systems                                | Mainly focuses on vulnerability demonstration and mitigation rather than detailed change localization | Supports raw-memory verification, backend comparison, and tamper evidence                  |
| Backend Application-Data Comparison        | Stores expected application/NDEF content and compares it with later reads         | Simple and suitable for application-level validation                                                   | May ignore unused raw memory, configuration areas, and security-state changes                         | Motivates verification beyond visible NDEF/application content                             |
| Flat Whole-Memory SHA-256                  | Generates a hash of the complete memory image and compares it with a trusted hash | Detects whether the current data differs from the trusted state                                        | Gives only MATCH/DIFFERENT; cannot identify the changed page, block, file, or field                   | Used as a baseline method for comparison with the proposed region-aware approach           |
| Password / Key / Access-Control Protection | Restricts unauthorised reading or writing using card-family security mechanisms   | Reduces unauthorised access where correctly configured                                                 | Does not explain what changed if protection is bypassed or authorised access is misused               | Supports separate security-state verification and configuration monitoring                 |
| Manufacturer / Chip                        | Verifies the origin or authenticity of                                            | Useful for checking                                                                                    | Does not prove that application or user data has remained                                             | Shows the difference between physical                                                      |

| Authenticity Mechanisms                                               | supported NFC chips                                                                                                                                    | supported chip provenance                                                  | unchanged after issuance                                                                  | authenticity and data-state integrity                                     |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Proposed NFC Tag Data Integrity Verification with Tamper Localization | Protected baseline, application/NDEF verification, raw-memory verification, security-state verification, change localization, and integrity journaling | Identifies what changed, where it changed, and the affected logical region | Requires trusted enrolment and cannot prove physical originality of a byte-identical copy | Addresses the identified integrity-verification and forensic-evidence gap |

*Table 1 Comparison of Existing NFC Tag Integrity Verification Approaches*

### 1.3 Research Gap

Strong evidence that contactless card systems are vulnerable to cloning, tampering, replay, insufficient access control settings, and other security issues may be found in existing NFC and RFID studies [1], [2]. Nevertheless, vulnerability detection by itself does not offer a practical way to ascertain whether the information currently on an enrolled NFC card is still the same as it was in its trusted condition. Specifically, empirical MIFARE Classic research shows that targeted memory modification is feasible in real world systems [2], although the primary focus of that study is vulnerability demonstration rather than precise post detection localization of altered card data.

The extent of current integrity mechanisms is a second drawback. Application level NDEF records are safeguarded by NDEF signature and authentication techniques, which can also confirm if signed message content has been changed [3]-[5]. Unused readable memory, lock information, configuration data, access conditions, or application/file security settings are examples of integrity relevant information that an NFC card may store outside of the displayed NDEF message. As a result, a successful NDEF verification does not guarantee that the appropriate card state has not changed. By dividing integrity verification into three categories application/NDEF verification, raw memory verification, and security state verification the proposed study fills this gap.

Tamper localization has an additional gap. A single whole-state digest simply shows whether the two states match, while SHA-256 offers a robust method for comparing a trusted state with a subsequent observation [12]. The field, page, block, file, byte range, or logical region that caused the discrepancy is not identified. Therefore, when a modification has been found, a straightforward flat hash comparison is not enough to support an investigation. In order to report the impacted place and previous/current values where available, the proposed work expands on simple hash comparison through region-aware and smaller-unit comparison.

Additionally, current methods only partially provide a common verification model across various NFC card architectures. The storage structures used by NTAG215, MIFARE Classic 1K, and MIFARE DESFire differ significantly [14]-[16]. While treating each DESFire card as a flat byte array loses significant structural and security meaning, a page oriented comparison method cannot

be easily applied to an application/file-oriented DESFire card. Therefore, while maintaining a single common verification procedure, the proposed research introduces family aware parsing, canonicalization, and logical region mapping.

The reliability of the baseline itself is another significant gap. An original, reliable reference is essential to any integrity system based on comparisons. A later compromised card state can mistakenly appear genuine if that reference is silently changed or replaced. While the secret key is kept safe, HMAC offers keyed authentication that may be used to identify unauthorized changes to a stored baseline [13]. Therefore, rather than assuming that the reference database is always reliable, the suggested research incorporates HMAC protected baseline manifests, baseline version control, and explicit handling of baseline integrity failures.

Additionally, a state transition traceability gap is addressed in the proposed approach. An authorized update or an unexplained alteration may not always be the cause of a discrepancy between the present card state and a previous baseline. In order to match supported card resident evidence with that trusted history, the proposed integrity journal logs authorized enrollment and update events in the backend. This enables the verification process to differentiate between unjournalled changes, conflicting journal evidence, and rollback situations from typical authorized transitions.

Lastly, effective NFC verification needs to differentiate between insufficient evidence and confirmed modification. Unreadable areas, unsupported access circumstances, or interrupted scans might cause physical card capture to fail or yield insufficient evidence. False modification findings may rise if all incomplete observations are treated as modifications. Therefore, when there is not enough data to justify a VALID or MODIFIED conclusion, the proposed system incorporates read-quality validation and an explicit UNKNOWN state.

| <b>Aspect</b>            | <b>Original Integrity Verification Method</b> | <b>Proposed NFC Tag Data Integrity Verification with Tamper Localization</b> |
|--------------------------|-----------------------------------------------|------------------------------------------------------------------------------|
| Main verification method | Whole-memory / application SHA-256 comparison | Multi-layer region-aware verification                                        |
| Application data         | Hash-based equality checking                  | NDEF/application comparison with field-level difference identification       |
| Raw memory               | Whole-memory hash                             | Page/block/file-level comparison and change localization                     |
| Security state           | Limited or not separately analysed            | Supported configuration, access, lock, and security-state verification       |
| Change location          | Not identified                                | Identifies page, block, file, field, or logical memory region                |

|                                    |                                                 |                                                                                                                                                    |
|------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Old / new values                   | Not normally reported                           | Recorded where available                                                                                                                           |
| Trusted baseline protection        | Basic stored reference                          | HMAC-protected baseline manifest                                                                                                                   |
| Duplicate enrolment                | May overwrite the trusted reference             | Existing baseline is checked and unintended re-enrolment is rejected                                                                               |
| Poor / incomplete read             | May appear as a mismatch                        | Read-quality failure can produce UNKNOWN                                                                                                           |
| Integrity history                  | No structured event history                     | Backend integrity journal with card-resident evidence where supported                                                                              |
| Authorised update handling         | New state may simply replace previous reference | Controlled update recorded as a new trusted event/version                                                                                          |
| Unknown external modification time | Not addressed                                   | Reports last known valid time and first detected modified time as a modification window                                                            |
| Card-family support                | Primarily one memory representation             | Family-aware support for NTAG215, MIFARE Classic 1K, and MIFARE DESFire                                                                            |
| Final output                       | VALID / MODIFIED / UNKNOWN                      | VALID / MODIFIED / UNKNOWN plus detailed tamper evidence                                                                                           |
| Evaluation focus                   | Demonstration of hash mismatch                  | Localization accuracy, classification accuracy, false-modification rate, journal verification, cross-reader consistency, UNKNOWN rate, and latency |

*Table 2 Comparison of the Original Integrity Verification Method and the Proposed Method*

## 2. OBJECTIVES

### 2.1 Main Objective

This research's primary goal is to develop, implement, and test an NFC tag data integrity verification technique that can detect and localize changes in application data, raw memory, and supported security state information while preserving tamper evident integrity event records and safeguarding the trusted baseline.

The suggested approach will determine whether an NFC card or tag's present state still corresponds to its authorized enrolled status and will yield one of three main results: VALID, MODIFIED, or

UNKNOWN. When a modification is detected, the method will try to identify the field, page, block, file, or logical memory region that is impacted. It will also offer supplementary evidence, such as journal consistency, pertinent timestamps, and prior and current values. Controlled laboratory tests using approved NFC test cards and established modification ground truth will be used to assess the study.

## 2.2 Specific Objectives

1. To **establish a family-aware memory and data-region model** for NTAG215, MIFARE Classic 1K, and MIFARE DESFire so that a consistent integrity-analysis procedure may be used to verify application data, raw memory, configuration information, and other supported security-related states.
2. To **create a trusted enrollment process** that records the authorized NFC card state, establishes a baseline with application/NDEF, raw memory, and security state evidence, uses HMAC to secure the stored baseline, and stops accidental duplicate enrollment from silently replacing an existing trusted reference.
3. To put in place a **three-layer integrity-verification system** that includes application/NDEF verification, raw memory verification, and security-state verification. This will enable a methodical comparison of the present NFC card state with the trusted baseline.
4. To provide a **tamper-localization and classification system** that, upon detection of a modification, determines the impacted field, page, block, file, or logical memory region and categorizes the event using a predetermined tamper taxonomy.
5. To enable the system to **differentiate authorized updates** from unjournalized changes, missing journal evidence, inconsistent journal state, and rollback conditions by implementing tamper-evident integrity-event recording using an independent backend journal and compact card-resident evidence when supported.
6. To measure tamper-localization accuracy, tamper-class classification accuracy, false-modification rate, journal-verification accuracy, cross-reader consistency, UNKNOWN rate, and verification latency against traditional baseline methods in an experimental setting using controlled modification scenarios with known ground truth.

## 3. METHODOLOGY

### 3.1 Research Methodology and Design

The suggested **NFC Tag Data Integrity Verification** with Tamper Localization component is designed, implemented, and assessed in this study using a controlled experimental technique. Because the system needs a known trusted NFC state, deliberate changes, and known ground-truth results for comparison, controlled testing is appropriate. The research team's NFC cards and tags

or those specifically approved for academic assessment will be the only ones used. Flipper Zero will be utilized for cross-reader checks and secondary validation, while Proxmark3 will serve as the main research and measurement tool.

Authorized enrollment and subsequent verification are the two primary phases of the approach. The original NFC card is scanned, its card family is determined, and pertinent application/NDEF data, readable raw memory, and supported security-state data are gathered during enrollment. The data is mapped into logical regions according to the card family after being canonicalized to eliminate superfluous formatting changes. After that, a trusted baseline is kept in the backend and secured by **HMAC**. The system maintains the various structures of **NTAG215**, **MIFARE Classic 1K**, and **MIFARE DESFire**, including pages, sectors/blocks, and applications/files.

The card is scanned once more during subsequent verification, and the fresh observation is considered the current state. The system first determines if sufficient trustworthy evidence has been gathered. Instead of inaccurately reporting a modification, the result is **UNKNOWN** if the scan is faulty or incomplete. The trusted baseline is obtained and its **HMAC** is confirmed prior to comparison if the scan is appropriate. Three tests are then carried out by the verification process: **security-state**, **raw-memory**, and **application/NDEF** verification.

Instead of just **reporting a generic mismatch**, the system does tamper localization if a difference is found. The impacted location may be characterized as an **NDEF field, byte range, page, block, sector, application, file, record, or logical area, depending on the card family**. When available, both current and past values are kept. While direct page, block, or file comparison is still a crucial benchmark, comparisons can be arranged using region level hashes or a hash tree structure.

In order to differentiate between authorized updates and inexplicable changes, the system additionally keeps an integrity-event journal. While compact card resident evidence may be employed when securely supported, the entire trusted event history is stored in the backend. A trusted timestamp can be stored by the system for authorized updates. It stores the Last Known Valid Time and First Detected Modified Time to indicate a potential modification interval rather than claiming the precise modification time for unknown external changes. Together with supporting documentation such the impacted layer, location, journal condition, timestamps, and modified values, the ultimate determination is either **VALID**, **MODIFIED**, or **UNKNOWN**. After each verification, the system generates and stores a structured verification log/report containing the final result, card family, scan ID, baseline version, affected verification layer, localized change, previous and current values where available, journal status, timestamps, reader information and verification latency.

For every supported **NFC family**, controlled, unaltered, and modified scenarios will be developed for assessment. It will be possible to compare the intended location and verdict with the system output by introducing known changes to the application/NDEF, **raw-memory**, or **supported security-state layers**. To determine whether typical reader variance resulted in erroneous **MODIFIED** results, repeated unaltered scans will also be utilized. By including tamper localization, integrity logging, three-layer verification, family-aware interpretation, protected baseline management, and explicit treatment of doubtful reads, the methodology expands upon the

previous straightforward SHA-256 comparison approach. Because the system uses deterministic comparison criteria and known trusted states, machine learning is not necessary.

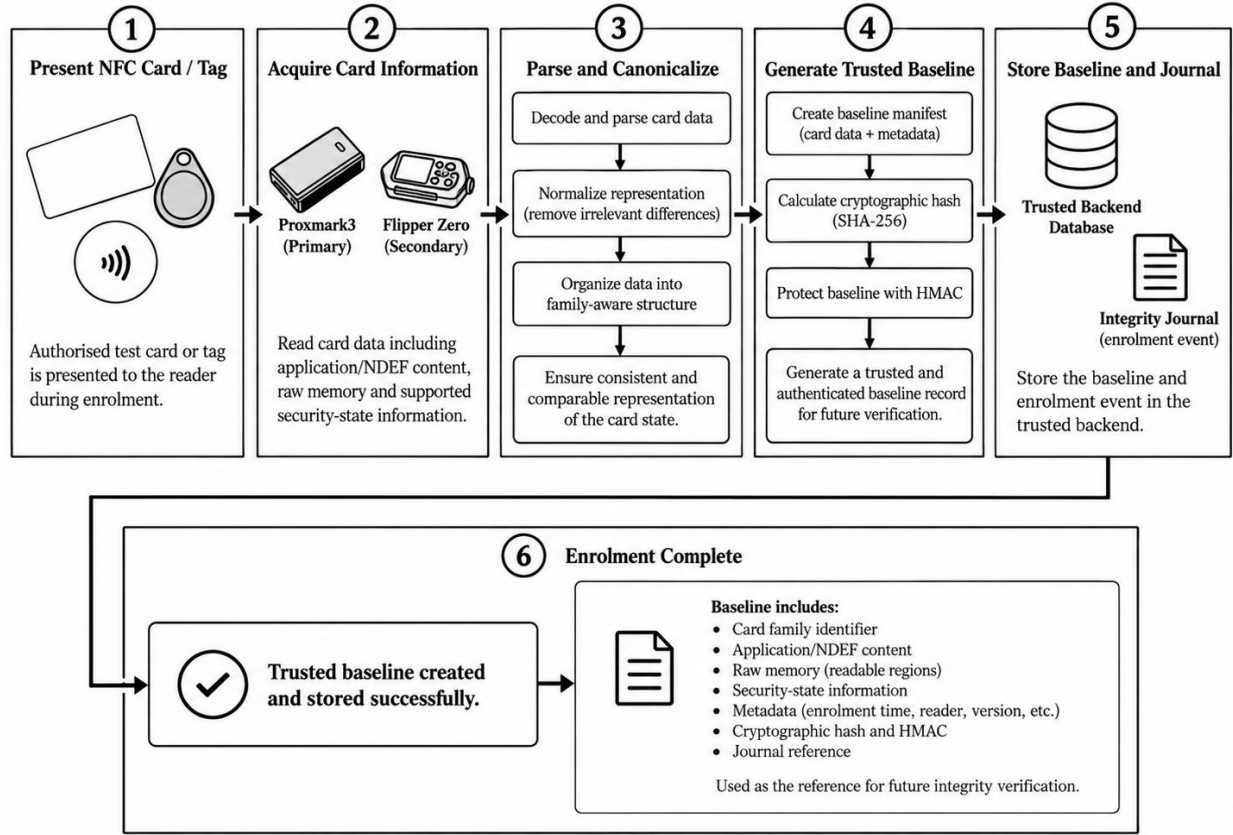


Figure 2 Authorised NFC Tag Enrolment and Trusted Baseline Creation Process

### 3.2 Data / Experimental Inputs

A controlled experimental dataset gathered from authorized **NTAG215**, **MIFARE Classic 1K**, and **MIFARE DESFire** cards will be used in the proposed NFC Tag Data Integrity Verification with Tamper Localization component. The initial trusted card state, subsequent current scans, controlled modified states, degraded or incomplete reads, and journal-related evidence are the primary inputs. Metadata including the card family, scan ID, reader used, baseline version, timestamp, expected ground truth, and final verification outcome will also be included in each trial. Flipper Zero will be utilized for specific secondary validation, and Proxmark3 will serve as the primary acquisition device.

Machine learning training will not make use of the dataset. Rather, it will facilitate deterministic comparison between the present NFC state and the trusted baseline. Application/NDEF data,

readable raw-memory data, supported security-state information, hashes, logical region mappings, and journal records are all examples of the stored evidence. In order to objectively assess the system's detection, classification, and localization accuracy, controlled adjustments will be documented together with the known affected location and anticipated outcome.

| <b>Characteristic</b>                             | <b>NTAG215</b>                                                                                | <b>MIFARE Classic 1K</b>                                                                | <b>MIFARE DESFire</b>                                                                      |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Primary data organisation                         | Page-oriented memory                                                                          | Sector- and block-oriented memory                                                       | Application- and file-oriented storage                                                     |
| Typical smallest physical/logical comparison unit | Page / byte range                                                                             | Block / byte range                                                                      | File, record, or byte range where readable                                                 |
| Application data                                  | Commonly NDEF/TLV data in user memory                                                         | Application-specific data stored in sector blocks                                       | Application-specific files and records                                                     |
| Raw-state verification                            | Readable pages and relevant bytes                                                             | Readable blocks and sectors                                                             | Readable application/file contents and supported metadata                                  |
| Security-related state                            | Lock/configuration and supported protection information                                       | Sector trailers, access conditions, and authentication-related state where available    | Application/file access rights and supported security configuration                        |
| Tamper-location output                            | Page, byte offset, field, or logical region                                                   | Sector, block, byte offset, or logical region                                           | Application, file, record/offset, or logical region                                        |
| Important reserved areas                          | Manufacturer/system and configuration areas must not be treated as general application memory | Manufacturer block must not be repurposed; sector trailers contain security information | Structure is managed through applications and files rather than arbitrary page/block reuse |
| Journal approach                                  | Compact evidence only in verified unused user memory where available                          | Dedicated authorised application sector/blocks where appropriate                        | Dedicated integrity-journal application/files where appropriately configured               |
| Full trusted history                              | Backend journal                                                                               | Backend journal                                                                         | Backend journal                                                                            |
| Role in this research                             | Page-level integrity and NDEF tamper localisation                                             | Sector/block integrity and access-state verification                                    | Application/file-oriented integrity and security-state verification                        |

*Table 3NFC Card Family Memory Region Comparison*

### 3.3 Baseline and Benchmark Methods

A reliable baseline and multiple benchmark verification techniques are used in the experimental assessment. While the benchmark methods offer more straightforward ways to compare the current state with that reference, the trusted baseline represents the authorized state of an NFC card or tag recorded during enrollment. The system gathers readable application/NDEF data, raw-memory data, and supporting security-state information during enrollment. It then canonicalizes the data to prevent innocuous formatting variations from producing incorrect modification results. Card-family information, baseline version, enrollment event ID, trustworthy timestamp, reader data, logical region mappings, and pertinent hashes or content proof are also stored in the baseline.

**HMAC** is used to secure the stored baseline, allowing for the detection of unauthorized changes to the trusted reference prior to verification. Rather than merely designating the card as changed, the system indicates a baseline-integrity issue if the HMAC check is unsuccessful. Additionally, baseline management is version aware. An authorized update must establish a new baseline version and document the occurrence in the backend journal while maintaining prior trusted states; a new card scan cannot silently replace an existing trusted baseline.

Three benchmark techniques will be compared with the suggested strategy. The first is **application/NDEF-only comparison**, which may overlook changes outside of the NDEF region and only examine the visible application content. The second is a **flat whole-state SHA-256 comparison**, which can identify if the entire state is the same or different but is unable to pinpoint the precise location of the difference. **Direct memory-unit comparison** is the third method; depending on the card family, it compares individual pages, blocks, files, records, or byte ranges. Because it can already give precise localization for modest NFC memory, this direct comparison is a crucial benchmark.

Three coordinated verification layers application/NDEF verification, **raw-memory verification**, and **security-state verification** are used in the suggested method to expand these benchmarks. For the system to determine whether there is a difference, as well as what changed and where, evidence from these layers is paired with **baseline authentication** and reliable journal data. Evidence may be arranged using hierarchical or Merkle-like hashing, but the research will not presume that this is inherently superior to direct comparison.

When the necessary trusted evidence matches the expected state, the outcome is **VALID**; when trustworthy evidence reveals an inexplicable integrity-relevant difference, it is **MODIFIED**; and when the available evidence is insufficient, untrustworthy, or unreadable, it is **UNKNOWN**. Acquisition failures are not mistakenly interpreted as tampering because to our three-state approach. The three verification layers, the evidence each layer examines, and the localization output generated when a difference is found are all summarized in Table 4.

| <b>Verification Layer</b>       | <b>Primary Evidence Examined</b>                                                                                     | <b>Example Detected Change</b>                                       | <b>Localization Output</b>                                                            | <b>Main Contribution</b>                                        |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Application / NDEF Verification | NDEF records, application fields, payloads, URIs, text and supported semantic content                                | URL changed, text payload modified, application field replaced       | Record, field, payload or semantic element                                            | Identifies visible/application-level modification               |
| Raw-Memory Verification         | Readable pages, blocks, files, byte ranges and unused readable memory                                                | Byte or memory unit changed while visible NDEF remains unchanged     | Page, block, file, offset, byte range or logical region                               | Detects modifications outside normal application representation |
| Security-State Verification     | Supported lock/configuration state, access conditions, sector/file access information and readable security metadata | Lock state altered, access condition changed, configuration modified | Security field, configuration region, sector trailer or file/application access state | Identifies integrity-relevant security/configuration changes    |
| Combined Evidence Output        | Results from all three layers plus baseline and journal evidence                                                     | One or more integrity-relevant differences                           | Affected layer + supported location + previous/current values where available         | Produces evidence-rich VALID, MODIFIED, or UNKNOWN decision     |

*Table 3 Multi-Layer Integrity Verification Layers and Output*

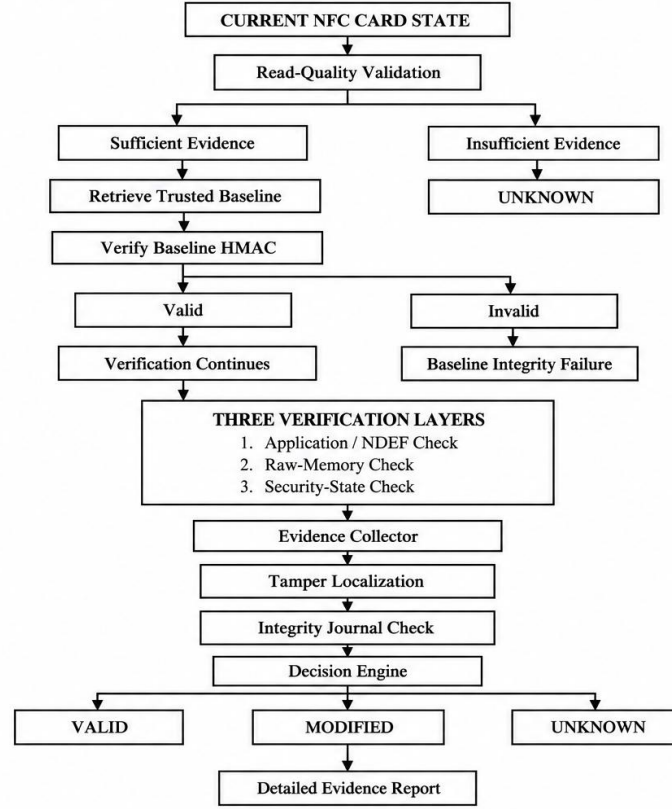


Figure 3 Three-Layer NFC Tag Integrity Verification Process

Consequently, the comparative study will examine more than just change detection. Each method will be evaluated according to how well it can recognize unmodified cards, identify controlled modifications, locate the known modification, provide enough explanations, and react to incomplete or incorrect current readings. The proposed method is expected to produce a richer integrity result than NDEF-only or flat-hash verification, even if direct memory-unit comparison is an essential reference for evaluating the added value of hierarchical and region-aware evidence processing.

### 3.4 Evaluation Metrics

Measures that evaluate decision accuracy, tamper localization, classification accuracy, dependability, journal verification, cross-reader consistency, and performance will be used to evaluate the proposed NFC Tag Data Integrity Verification with Tamper Localization component. Every verification outcome shall be contrasted with established experimental ground truth. The assessment will verify that unaltered cards are appropriately reported as **VALID**, controlled modifications as **MODIFIED**, and insufficient or faulty evidence as **UNKNOWN**. Additionally, it will check if the system accurately pinpoints the precise location of a detected change and the impacted layer. The evaluation will additionally measure the **false-modification rate**, **UNKNOWN rate**, evidence completeness, cross-reader consistency, integrity-journal accuracy,

and verification delay. The primary measurement evidence will come from Proxmark3, and some secondary validation will be done using Flipper Zero. To ascertain whether the multi-layer strategy offers superior localization and more valuable verification evidence, the suggested method will also be compared with application/NDEF-only comparison, flat whole-state SHA-256 comparison, and direct memory-unit comparison.

| <b>Evaluation Metric</b>             | <b>Measurement Purpose</b>                                                                             | <b>Ground Truth / Reference</b>                              | <b>Interpretation</b>                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| Verdict Correctness                  | Determines whether the final VALID, MODIFIED, or UNKNOWN decision matches the known test condition     | Controlled experimental ground truth                         | Higher agreement indicates better decision correctness        |
| Tamper-Localization Accuracy         | Determines whether the reported modification location matches the actual modified location             | Recorded page, block, file, field, offset, or logical region | Higher accuracy indicates better localization capability      |
| Tamper-Class Classification Accuracy | Determines whether the detected change is assigned to the correct modification category                | Predefined experimental tamper class                         | Higher accuracy indicates better explanatory capability       |
| False-Modification Rate              | Measures unchanged cards incorrectly classified as MODIFIED                                            | Repeated unchanged-card controls                             | Lower values indicate greater reliability                     |
| UNKNOWN Rate                         | Measures the frequency of cases where sufficient trusted evidence is unavailable                       | Verification results and scan-quality records                | Should remain low under normal acquisition conditions         |
| Journal Verification Accuracy        | Measures correct identification of authorised, unjournalled, inconsistent, or rollback states          | Controlled journal scenarios                                 | Higher agreement indicates reliable journal verification      |
| Cross-Reader Consistency             | Determines whether equivalent observations from different readers produce consistent integrity results | Proxmark3 and selected Flipper Zero observations             | Higher consistency indicates reader-independent verification  |
| Verification Latency                 | Measures the time required to produce the final integrity result and evidence                          | Recorded processing time                                     | Lower practical latency is desirable                          |
| Evidence Completeness                | Measures whether available location, values, layer, journal                                            | Expected evidence structure                                  | Greater completeness improves explainability and auditability |

|  |                                                 |  |  |
|--|-------------------------------------------------|--|--|
|  | status, and supporting information are reported |  |  |
|--|-------------------------------------------------|--|--|

*Table 4 Evaluation Metrics and Measurement Criteria*

### 3.5 Validation Strategy

Authorized NTAG215, MIFARE Classic 1K, and MIFARE DESFire cards will be used in controlled laboratory trials to evaluate the proposed NFC Tag Data Integrity Verification with Tamper Localization component. To establish a reliable baseline, each card will first be enrolled. After that, the cards will be examined under journal-related, degraded-read, controlled-modification, and unaltered situations. The system output and the experimental ground truth can be readily compared because the expected condition and modification site are known before each experiment.

Validation will ascertain whether the system successfully identifies the impacted verification layer and modification location as well as whether it generates VALID, MODIFIED, or UNKNOWN results. Flipper Zero will be utilized for specific secondary observations and cross-reader consistency checks, while Proxmark3 will serve as the main acquisition and measurement tool. Additionally, repeated unaltered-card scans will be utilized to ascertain whether typical reader variance or inaccurate acquisition results in MODIFIED

| Test Scenario                        | Expected Result | Main Validation Purpose                       |
|--------------------------------------|-----------------|-----------------------------------------------|
| Unchanged enrolled card              | VALID           | Verify normal integrity checking              |
| NDEF/application data changed        | MODIFIED        | Verify application-layer detection            |
| Raw-memory region changed            | MODIFIED        | Verify raw-memory detection                   |
| Supported security-state changed     | MODIFIED        | Verify security-state detection               |
| Authorised update with valid journal | VALID           | Verify trusted state transition               |
| Change without trusted journal event | MODIFIED        | Detect unjournalled modification              |
| Journal inconsistency or rollback    | MODIFIED        | Verify journal validation                     |
| Incomplete or unreliable scan        | UNKNOWN         | Verify safe handling of insufficient evidence |

*Table 5 Controlled Modification Scenarios and Expected Results*

#### 3.5.1 Tamper Classification Validation

Any detected changes will be categorized using a predefined tamper taxonomy. By defining the taxonomy before examining the final experimental data, it is possible to avoid developing categories only after viewing the system output. The categorization technique will primarily describe the type of integrity evidence affected, rather than trying to infer an attacker's purpose.

| <b>Tamper Class</b>             | <b>Description</b>                                                                           | <b>Typical Evidence</b>                                          | <b>Example</b>                                   |
|---------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------|
| Application / NDEF Modification | Change to decoded application-level or NDEF information                                      | Record, field, payload, URI or text difference                   | Authorized URL replaced                          |
| Raw-Memory Modification         | Difference in readable underlying memory                                                     | Page, block, file, offset or byte-range difference               | Unused writable memory changed                   |
| Security-State Modification     | Difference in supported security/configuration state                                         | Lock, configuration, access-condition or access-right difference | Access condition altered                         |
| Unjournalled Change             | Current state differs without a corresponding trusted update event                           | State difference + no matching backend event                     | External rewrite after last trusted verification |
| Journal Inconsistency           | Card-resident evidence and trusted backend history disagree                                  | Sequence/version/event mismatch                                  | Card journal value does not match backend        |
| Rollback                        | Current evidence corresponds to an older trusted state rather than the latest accepted state | Older baseline/journal version detected                          | Previous valid state restored                    |
| Incomplete Evidence             | Required evidence cannot be acquired reliably                                                | Missing/unreadable region or failed scan                         | Interrupted NFC acquisition                      |
| Baseline Integrity Failure      | Stored trusted baseline cannot be authenticated                                              | HMAC verification failure                                        | Backend baseline altered without valid HMAC      |

*Table 6 Tamper Classification Taxonomy*

### 3.5.2 Tamper-Localization Validation

Finding out if the system accurately detects the location of a modification is a significant portion of the validation. The known location noted during controlled alteration will be compared to the reported location. Localization can identify an NDEF field, byte range, NTAG215 page, MIFARE Classic sector or block, or MIFARE DESFire program, file, or record, depending on the NFC family.

Since direct page, block, file, or byte-level comparison can already offer precise localization for modest NFC memory, it will be utilized as a crucial benchmark. For the purpose of organizing region-level integrity evidence, a hierarchical or Merkle-like hash structure may also be assessed; however, the research will not presume that it is inherently more accurate than direct comparison.

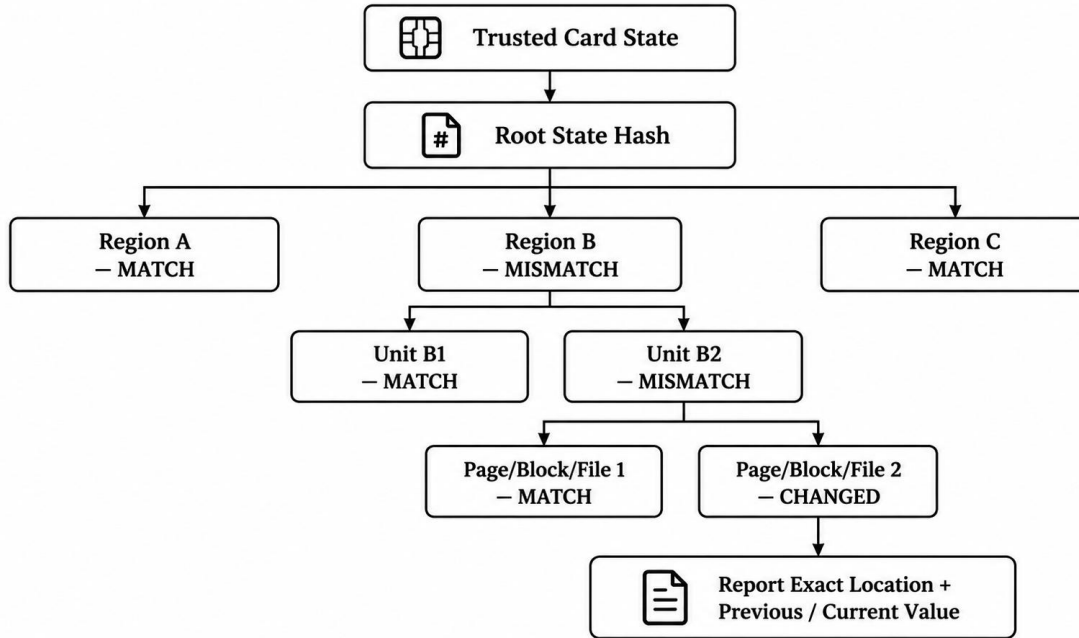


Figure 4 Hash-Tree-Based Memory Change Localization

### 3.5.3 Experimental Sample and Verification Read Distribution

Nine physical test cards total—three from each supported NFC family will be used in the controlled laboratory experiment. These cards will undergo several controlled trials to assess degraded-read circumstances, purposeful alterations, and unaltered states.

| NFC Card Family   | Physical Test Cards | Unchanged Verification Trials | Controlled Modification Trials | Incomplete / Degraded Read Trials | Total Primary Trials |
|-------------------|---------------------|-------------------------------|--------------------------------|-----------------------------------|----------------------|
| NTAG215           | 3                   | 30                            | 30                             | 15                                | 75                   |
| MIFARE Classic 1K | 3                   | 30                            | 30                             | 15                                | 75                   |
| MIFARE DESFire    | 3                   | 30                            | 30                             | 15                                | 75                   |
| <b>Total</b>      | <b>9</b>            | <b>90</b>                     | <b>90</b>                      | <b>45</b>                         | <b>225</b>           |

Table 7 Experimental Sample and Verification Read Distribution

### 3.5.4 Trusted Baseline Security Validation

To ascertain whether the system can identify issues impacting the integrity of the stored reference itself, the trusted baseline will also be examined. Baseline-content change, inaccurate HMAC

values, unauthorized baseline replacement, repetitive re-enrollment attempts, baseline rollback, and journal alteration are examples of controlled tests.

Instead of immediately labeling the NFC card as MODIFIED, the system should report a trusted-reference integrity failure if the baseline HMAC fails. Because the issue can be with the stored reference rather than the actual card, this distinction is crucial.

| Threat or Failure Condition                      | Expected Protection / System Response                                      | Residual Limitation                                                              |
|--------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Baseline data modified without the HMAC key      | HMAC verification fails and the baseline is rejected                       | Depends on correct implementation and protection of the HMAC mechanism           |
| Baseline and ordinary SHA-256 hash both replaced | HMAC authentication should still fail without the secret key               | Compromise of the HMAC key can defeat this protection                            |
| Modified card presented for normal re-enrolment  | Existing trusted record is detected and unintended replacement is rejected | Legitimate administrative override must be securely controlled                   |
| Older trusted baseline restored                  | Baseline version and journal history are checked for rollback              | Requires trusted version and journal records to remain available                 |
| Backend journal entry removed or altered         | Journal inconsistency should be identified during validation               | Complete backend compromise may bypass software-only controls                    |
| HMAC key disclosed                               | An attacker may generate valid HMAC values for modified baselines          | Key compromise is outside the protection guaranteed by this prototype            |
| Verification host fully compromised              | Integrity results are no longer considered trustworthy                     | Requires stronger operational or hardware security beyond the research prototype |

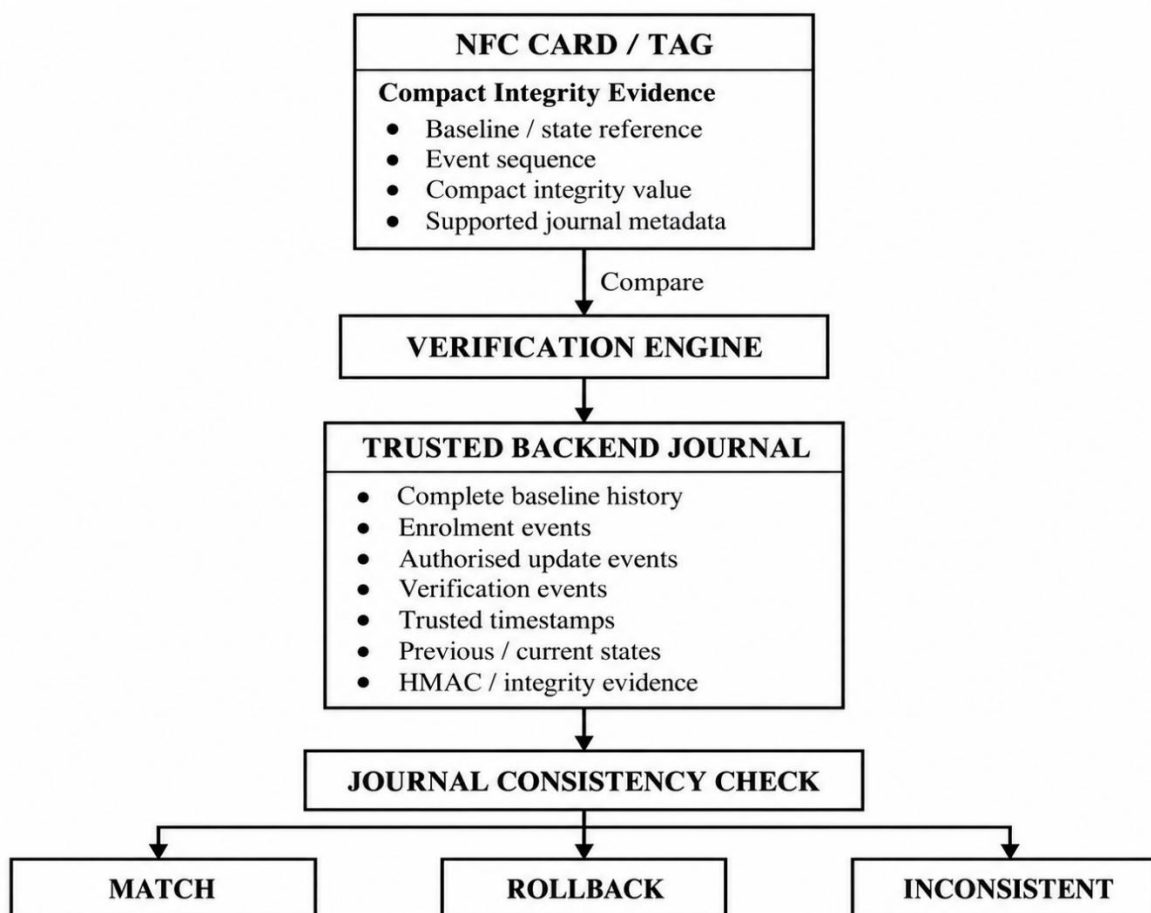
*Table 7 Trusted Baseline Threat Mode*

### 3.6 Development Approach

A modular and progressive approach will be used in the development of the suggested NFC Tag Data Integrity Verification with Tamper Localization component. NFC data collecting, card-family identification, parsing and canonicalization, trusted baseline management, HMAC verification, three-layer integrity checking, tamper localization, integrity-journal validation, decision generation, and reporting will all be handled by distinct modules inside the system. Without completely rebuilding the system, this structure makes it easy to accommodate **NTAG215**, **MIFARE Classic 1K**, and **MIFARE DESFire** and permits independent testing of each module.

Using **Proxmark3** as the main research tool and **Flipper Zero** as a supplementary validation tool, development will start with NFC data collection. After that, the gathered data will be processed, canonicalized, and mapped into logical regions unique to each card family. The three verification layers application/NDEF, raw memory, and security-state verification will then be put into place after the trusted baseline and HMAC protection methods. The impacted page, block, sector, application, file, record, byte range, or logical region, if supported, will then be identified by the tamper-localization module.

integrity-journal subsystem will also be part of the prototype. The complete trusted history of enrollment and authorized updates will be stored on the backend, and where the card family safely permits it, compact card-resident evidence may be utilized. Authorized changes can be distinguished from unjournaled alterations, rollback situations, and inconsistent histories with the use of journal information.



*Figure 5 Card-Resident and Backend Integrity Journal Architecture*

The verification, localization, journal, and reporting modules will all be integrated into a single workflow during the last stages of development. Along with supporting data, the decision engine will generate VALID, MODIFIED, or UNKNOWN results. Accuracy, localization, false-modification behavior, journal handling, cross-reader consistency, and verification latency will next be assessed using controlled tests. Incremental development will allow for the validation of each stage prior to the integration of the subsequent module.

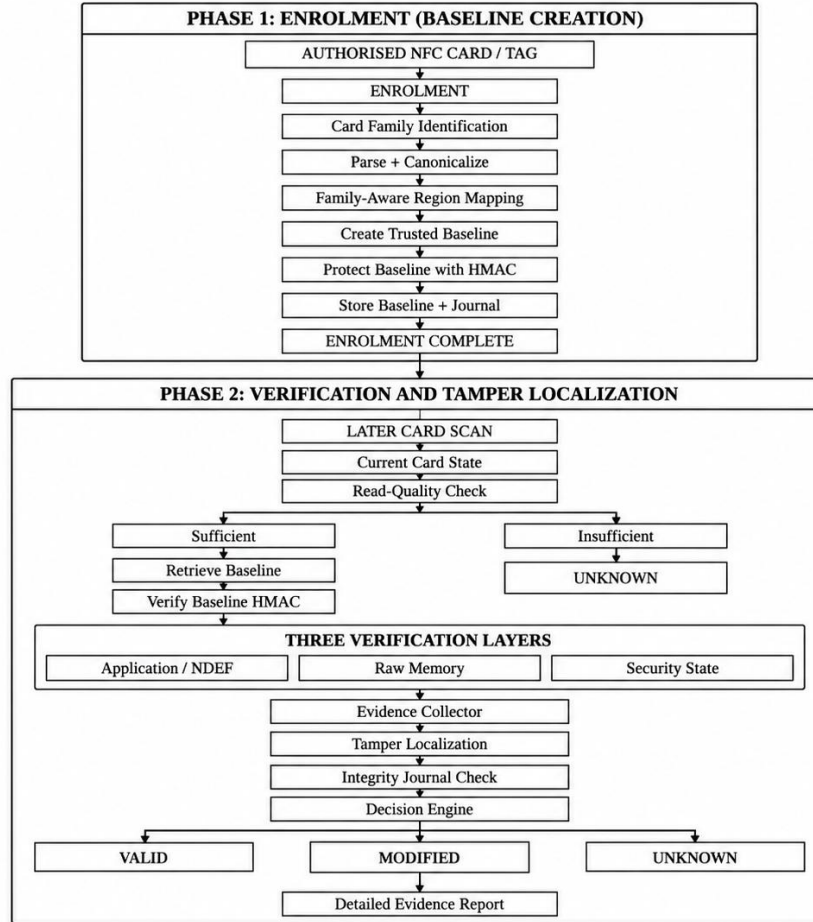


Figure 6 Complete NFC Tag Data Integrity Verification and Tamper Localization Workflow

## 4. HIGH-LEVEL SYSTEM ARCHITECTURE

### 4.1 System Architecture Overview

The proposed research introduces an NFC Card and Tag Security Enhancement Framework Using Hybrid Detection that combines four independent but complementary components: **C1 - NFC Security Profile Scoring**, **C2 - Multi-Session Protocol-Response Provenance Fingerprinting**, **C3 - NFC Tag Data Integrity Verification**, and **C4 - Hybrid Behavioral Anomaly Detection**. Each component addresses a different NFC security problem while contributing to a unified assessment.

The framework begins with authorized NFC cards and tags, with data collected through a shared acquisition layer using devices such as Proxmark3 RDV2, Flipper Zero, and ACR122U. After preprocessing, the collected evidence is provided to the four components in parallel, where each component performs its analysis independently and produces a component-specific result.

Finally, the system stores the outputs in a shared backend and presents them through a unified dashboard and reporting layer. This architecture maintains component independence while providing an integrated view of NFC card and tag security.

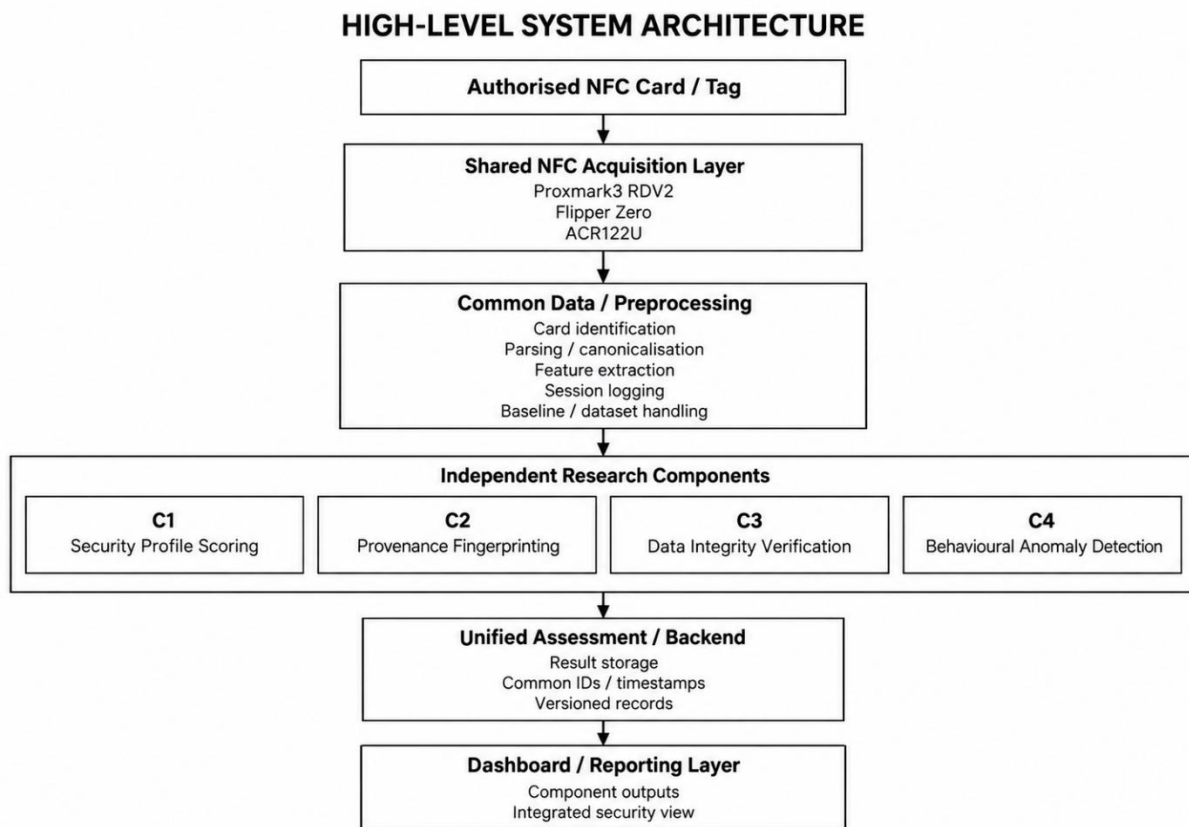


Figure 7 High-Level Architecture of the NFC Card and Tag Security Enhancement Framework

## 4.2 Architecture of the Individual Integrity Verification Component

Every NFC observation is processed through an organized verification pipeline by the suggested NFC Tag Data Integrity Verification with Tamper Localization component. The system first recognizes the card family after receiving the NFC scan. After that, the data is mapped into the appropriate logical regions for **NTAG215**, **MIFARE Classic 1K**, or **MIFARE DESFire** after

being parsed and canonicalized. Before verification proceeds, a read-quality check is also carried out to make sure there is sufficient trustworthy evidence.

The system then obtains the accurate trusted baseline and confirms its **HMAC**. Three levels are then used to verify the present NFC state: **application/NDEF** verification, raw-memory verification, and security-state verification. When a discrepancy is found, the tamper-localization module and evidence collector determine which field, page, block, sector, application, file, record, or logical region is impacted. Additionally, the integrity-journal module verifies that the authorized update history and the observed state match.

Finally, the decision engine integrates the information at hand and generates one of three outcomes: UNKNOWN, MODIFIED, or VALID. Additionally, the system creates a comprehensive verification report that includes the impacted layer, the location of the update, the journal condition, timestamps, baseline version, reader information, and prior and current values when available.

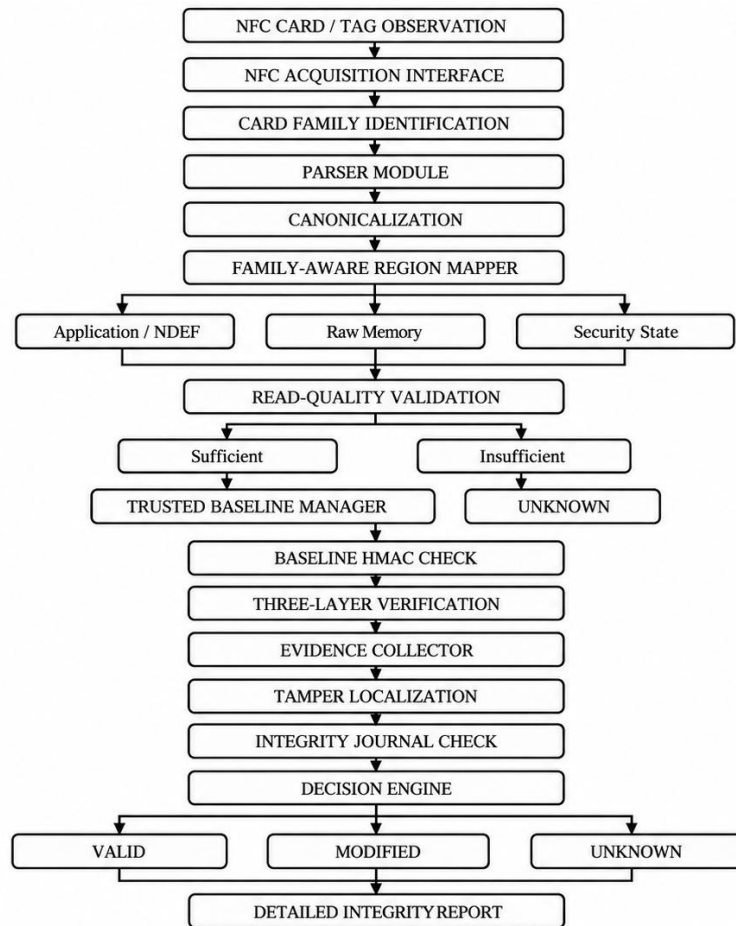


Figure 8 Internal Architecture of the Proposed Integrity Verification Component

### 4.3 Overall System Integration

Through a shared acquisition, backend, and reporting architecture, the suggested **NFC Tag Data Integrity Verification with Tamper Localization component** is incorporated into the larger NFC Card and Tag Security Enhancement Framework Using Hybrid Detection. The various study components receive NFC evidence obtained by the secondary Flipper Zero validation method or Proxmark3. After that, each component independently completes its own analysis and transmits the results to the shared backend.

The integrity verification component is independent of the outcomes of the behavioral analysis, security scoring, and clone-detection components. It generates **VALID, MODIFIED, or UNKNOWN together with tamper-localization and journal evidence** after independently assessing the stored NFC data state. To integrate and present the findings via the project dashboard, the shared backend uses common cards, scan, and event identifiers to store the outputs from all components.

This integration technique allows the entire framework to deliver a consolidated NFC security view while maintaining a clear separation of responsibility for each research component. As a result, a component's failure or unknown outcome does not necessarily alter the outcome that another component produces.

### 4.4 Individual Responsibility Boundary

Verifying if the stored data state of an enrolled NFC card or tag still matches its trusted baseline is the only task assigned to the proposed **NFC Tag Data Integrity Verification with Tamper Localization component**. The development of trusted baselines, HMAC protection, application/NDEF verification, raw-memory verification, security-state verification, tamper localization, integrity-journal checking, and the generation of VALID, MODIFIED, or UNKNOWN findings are some of its primary capabilities.

Physical clone or provenance detection, behavioral-anomaly detection, and overall numerical security grading are not performed by this component. These functions are part of the larger framework. This division guaranties that the integrity-verification component only makes assertions backed by the stored NFC data evidence and that each research component stays autonomous.

| Area / Function            | Within Individual Research Scope                                                      | Outside Individual Research Scope / Other Component                    |
|----------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| NFC card-state acquisition | Acquire or process authorized NDEF, raw-memory, security-state, and relevant metadata | Full physical fingerprint analysis belongs to clone-detection research |

|                               |                                                                                                                                  |                                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Card-family handling          | Support family-aware processing for NTAG215, MIFARE Classic 1K, and MIFARE DESFire                                               | Development of a universal parser for every NFC technology                                 |
| Parsing and canonicalization  | Convert supported reader/card evidence into a stable logical representation                                                      | Behavioral event-log feature engineering                                                   |
| Trusted baseline creation     | Create and maintain trusted enrolment state                                                                                      | Security posture scoring                                                                   |
| Baseline protection           | Authenticate trusted baseline using HMAC and manage versions                                                                     | Enterprise-wide key-management infrastructure                                              |
| Application/NDEF verification | Detect and localize supported application/NDEF changes                                                                           | General application business-logic validation unrelated to integrity                       |
| Raw-memory verification       | Detect and localize readable page/block/file-level changes                                                                       | Physical RF fingerprint verification                                                       |
| Security-state verification   | Compare supported lock, configuration, access-condition, and access-right state                                                  | Generate a general 0–100 NFC risk score                                                    |
| Tamper localization           | Identify affected field, page, block, file, offset, or logical region                                                            | Identify the physical origin of a cloned card                                              |
| Integrity journal             | Maintain and validate backend history and supported card-resident evidence                                                       | Guarantee an append-only card log against a fully privileged attacker                      |
| Authorised updates            | Record trusted state transitions and new baseline versions                                                                       | Recover exact time of an unobserved external modification                                  |
| Timestamp evidence            | Record trusted system-controlled event time and modification window                                                              | Claim exact external attack/modification time when it was not observed                     |
| Decision output               | Produce VALID, MODIFIED, or UNKNOWN with supporting evidence                                                                     | Produce clone classification, behavioral alert, or overall security score                  |
| Clone detection               | Data-state integrity may reveal changed copied content                                                                           | Byte-identical physical clone detection belongs to the separate clone/provenance component |
| Behavioral analysis           | None beyond integrity-event records required by this component                                                                   | Abnormal tap-pattern and access-behavior detection                                         |
| Experimental validation       | Evaluate localization, classification, false-modification behavior, journal verification, consistency, UNKNOWN rate, and latency | Evaluation metrics belonging exclusively to the other three research components            |
| Reporting integration         | Expose structured integrity results to the common dashboard                                                                      | Merge all independent results into a new integrity score                                   |

*Table 8 Individual Responsibility Boundary*

## 5. USER REQUIREMENTS

### 5.1 Requirements Elicitation and Stakeholder Needs

The user requirements for the proposed NFC Tag Data Integrity Verification with Tamper Localization component are based on the needs of security administrators, researchers, system operators, asset managers, and auditors who need to determine whether NFC card or tag data still matches an authorized trusted state. The system should provide trusted enrollment, clear VALID, MODIFIED, or UNKNOWN outcomes, and subsequent integrity verification. When a change is found, it should specify what was changed, where it occurred, the kind of data affected, and, if available, the previous and present values. It should also use HMAC to protect the trusted baseline, maintain integrity-journal records and baseline versions, and provide time-related evidence such as the last known valid time and the first detected modified time.

The system must also be reliable, understandable, and suitable for laboratory testing. It should not report incomplete or unreliable reads as confirmed modifications and instead return UNKNOWN when there is insufficient evidence. The same verification process should support NTAG215, MIFARE Classic 1K, and MIFARE DESFire while preserving their unique memory architectures. The integrity component should be limited to stored NFC data-state verification because behavioral anomaly detection, physical clone identification, and overall security score are separate parts of the larger architecture.

### 5.2 Functional Requirements

The primary functions that the suggested NFC Tag Data Integrity Verification with Tamper Localization component must carry out are specified in the functional requirements. NFC acquisition, trusted enrollment, baseline protection, tamper localization, integrity-journal checking, three-layer integrity verification, decision-making, and evidence reporting are all included in these standards.

| ID    | Functional Requirement             | Expected System Behavior                                                                                                            | Priority |
|-------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | Identify supported NFC card family | Determine whether the observed card is supported as NTAG215, MIFARE Classic 1K, or MIFARE DESFire and apply the appropriate profile | High     |
| FR-02 | Acquire authorised card state      | Capture the relevant readable application/NDEF, raw-memory, security-state, and acquisition metadata                                | High     |
| FR-03 | Parse and canonicalize evidence    | Convert reader/card evidence into a stable family-aware internal representation                                                     | High     |
| FR-04 | Create trusted baseline            | Generate and store the authorised enrolled card-state reference                                                                     | High     |
| FR-05 | Protect baseline using HMAC        | Authenticate the baseline manifest before future use                                                                                | High     |

|       |                                           |                                                                                                                              |        |
|-------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|
| FR-06 | Manage baseline versions                  | Preserve baseline history and prevent unintended silent replacement                                                          | High   |
| FR-07 | Validate current scan quality             | Determine whether sufficient reliable evidence exists for verification                                                       | High   |
| FR-08 | Retrieve trusted baseline                 | Locate the correct trusted baseline and latest accepted version                                                              | High   |
| FR-09 | Verify baseline HMAC                      | Reject or flag a baseline whose authentication fails                                                                         | High   |
| FR-10 | Verify application/NDEF data              | Compare structured application content and identify supported semantic changes                                               | High   |
| FR-11 | Verify raw memory                         | Compare readable family-specific memory or file content independently of NDEF                                                | High   |
| FR-12 | Verify security state                     | Compare supported lock, configuration, access-condition, or access-right information                                         | High   |
| FR-13 | Localize modifications                    | Identify affected field, page, block, sector, file, record, offset, or logical region                                        | High   |
| FR-14 | Classify integrity event                  | Assign the detected condition to the predefined integrity/tamper taxonomy                                                    | High   |
| FR-15 | Record previous and current values        | Preserve old/new values where reliable evidence is available                                                                 | Medium |
| FR-16 | Maintain backend integrity journal        | Record enrolment, authorized updates, verification events, versions, and trusted timestamps                                  | High   |
| FR-17 | Compare card and backend journal evidence | Identify matching, inconsistent, unjournaled, or rollback conditions where card evidence is supported                        | High   |
| FR-18 | Process authorized updates                | Create and authenticate a new trusted version after a legitimate state change                                                | High   |
| FR-19 | Produce VALID result                      | Report VALID when required trusted evidence matches the accepted state                                                       | High   |
| FR-20 | Produce MODIFIED result                   | Report MODIFIED when reliable integrity-relevant differences are confirmed                                                   | High   |
| FR-21 | Produce UNKNOWN result                    | Report UNKNOWN when sufficient trustworthy evidence is unavailable                                                           | High   |
| FR-22 | Generate detailed integrity report        | Present the verdict together with location, classification, journal status, values, metadata, and timestamps where available | High   |
| FR-23 | Record modification window                | Store last known valid and first detected modified times for unobserved external changes where available                     | Medium |

|       |                                    |                                                                                                         |        |
|-------|------------------------------------|---------------------------------------------------------------------------------------------------------|--------|
| FR-24 | Support cross-reader normalization | Allow equivalent supported observations from Proxmark3 and Flipper Zero to be normalized for comparison | Medium |
| FR-25 | Export structured result           | Provide component output in a format suitable for backend storage and dashboard integration             | Medium |
| FR-26 | Preserve component independence    | Operate without requiring clone-detection, security-scoring, or behavioral-analysis results             | High   |

*Table 9 Functional Requirements*

### 5.3 Non-Functional Requirements

The quality attributes anticipated from the suggested NFC Tag Data Integrity Verification with Tamper Localization components are outlined in the non-functional criteria. Reliability, security, performance, usability, interoperability, maintainability, reproducibility, auditability, and safe operation are the major concerns of these specifications.

| ID     | Non-Functional Requirement    | Expected Quality / Constraint                                                                  | Verification Method                                           |
|--------|-------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| NFR-01 | Reliability                   | Equivalent trusted inputs should produce consistent integrity decisions                        | Repeated unchanged and repeated-state testing                 |
| NFR-02 | False-modification resistance | Acquisition problems should not be reported automatically as confirmed modifications           | Unchanged and degraded-read experiments                       |
| NFR-03 | Baseline security             | Stored baselines must be HMAC-authenticated before use                                         | Controlled baseline-tampering tests                           |
| NFR-04 | Safe failure behaviour        | Insufficient, missing, or untrusted evidence must not produce an unsupported trusted result    | Incomplete-read, missing-baseline, and HMAC-failure scenarios |
| NFR-05 | Performance                   | Verification and evidence generation must complete within a practical laboratory workflow      | Verification-latency measurement                              |
| NFR-06 | Explainability                | Every non-trivial result should contain sufficient supporting evidence to explain the decision | Inspection of generated integrity reports                     |

|        |                            |                                                                                                                                |                                                |
|--------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| NFR-07 | Usability                  | VALID, MODIFIED, and UNKNOWN outcomes and their reasons should be clearly understandable                                       | Prototype/interface review                     |
| NFR-08 | Maintainability            | Acquisition, parsing, verification, localization, journal, and reporting functions should remain modular                       | Code/module architecture review                |
| NFR-09 | Cross-family compatibility | The logical workflow should support NTAG215, MIFARE Classic 1K, and MIFARE DESFire while preserving family-specific structures | Card-family validation tests                   |
| NFR-10 | Cross-reader consistency   | Equivalent supported observations should lead to consistent normalized verification results                                    | Proxmark3 and selected Flipper Zero comparison |
| NFR-11 | Reproducibility            | Experimental scenarios and outputs should be repeatable using documented test conditions                                       | Repeated controlled experiments                |
| NFR-12 | Traceability               | Each result should be linked to its baseline version, scan/test ID, evidence, and ground truth                                 | Experimental-record inspection                 |
| NFR-13 | Auditability               | Previous trusted states and integrity events should remain reviewable                                                          | Journal and baseline-history validation        |
| NFR-14 | History preservation       | Authorized updates should create versioned trusted states rather than silently overwrite history                               | Baseline-version tests                         |
| NFR-15 | Data privacy               | Unnecessary personal identifiers should not be collected or exposed                                                            | Dataset and report review                      |
| NFR-16 | Ethical operation          | Testing must be limited to owned or explicitly authorized NFC cards and tags                                                   | Research procedure and ethics documentation    |
| NFR-17 | Component independence     | Integrity verification must operate without depending on clone-detection, scoring, or behavioural-analysis outputs             | Independent component testing                  |

|        |                        |                                                                                                    |                     |
|--------|------------------------|----------------------------------------------------------------------------------------------------|---------------------|
| NFR-18 | Structured integration | Results should be exportable in a consistent format for backend storage and dashboard presentation | Integration testing |
|--------|------------------------|----------------------------------------------------------------------------------------------------|---------------------|

*Table 10 Non-Functional Requirements*

## 6. COMMERCIALIZATION PLAN

As an **NFC security and integrity-monitoring system**, the suggested NFC Tag Data Integrity Verification with Tamper Localization component offers commercialization potential. Security administrators, university IT teams, asset-management systems, NFC solution suppliers, and cybersecurity labs can all benefit from it. The system's primary benefit is that it not only indicates whether NFC data has changed, but also what has changed and where. It offers unambiguous outcomes that are VALID, MODIFIED, or UNKNOWN together with supporting data.

The system may be made available as an NFC security assessment tool, an **on premise software application, or an API/module** that may be included into current NFC management systems. Software licensing, enterprise subscriptions, NFC security assessment services, integration support, and technical maintenance are examples of potential commercial services. Future iterations may enable more compatible NFC readers, however Proxmark3 can be utilized for research and specialized testing.

After the research prototype has been thoroughly tested and verified, commercial deployment should start. Stronger key management, user access control, secure storage, backup, and audit logging would be necessary for a production version. Since physical clone detection is handled individually in the larger framework, the product should be sold as a defensive **NFC data-integrity and auditing solution rather than as a universal clone-detection system**.

## 7. BUDGET AND JUSTIFICATION

### 7.1 Hardware Costs

A limited set of NFC research hardware, test cards, and software development tools are needed for the planned study. To cut costs, free or open-source software and project resources already in place would be utilized whenever feasible. The authorized NFC test-card set needed for controlled laboratory studies and the NFC research equipment are the primary direct costs.

| Component    | Description                                                                                                                                                  | Estimated Cost (LKR) |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Flipper Zero | Secondary validation device used for supported NFC observations, NDEF inspection, selected cross-reader checks, and practical verification of test scenarios | 80,000               |
| Proxmark3    | Primary NFC research and acquisition device used for card identification, memory/NDEF acquisition, baseline creation,                                        | 25,000               |

|                                      |                                                                                                                                                                                                                                      |                |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|                                      | raw-memory analysis, and controlled integrity-verification experiments                                                                                                                                                               |                |
| NFC Card Set                         | Authorized test-card set containing the NFC card families required for this research, including NTAG215, MIFARE Classic 1K, and MIFARE DESFire cards for controlled enrolment, unchanged-state testing, and modification experiments | 10,000         |
| <b>Total Estimated Hardware Cost</b> |                                                                                                                                                                                                                                      | <b>115,000</b> |

*Table 11 Hardware Budget*

## 7.2 Software Tools and Development Platforms

Most software required for this research is freely available or open source. Therefore, no additional software license cost is expected for the prototype development stage.

| <b>Software / Tool</b>              | <b>Purpose</b>                                                                                                                                                                       | <b>Cost</b> |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Python 3.x                          | Main development language for parsing NFC data, canonicalization, SHA-256 hashing, HMAC validation, comparison logic, tamper localization, journal processing, and result generation | Free        |
| Proxmark3 Client / Iceman Firmware  | Controls Proxmark3 hardware and supports authorized NFC card acquisition, memory inspection, and laboratory analysis                                                                 | Free        |
| Visual Studio Code / Equivalent IDE | Source-code development, debugging, and project management                                                                                                                           | Free        |
| Git / GitHub                        | Source-code version control and development history management                                                                                                                       | Free        |
| SQLite / Open-Source Database       | Stores trusted baselines, baseline versions, integrity-journal records, verification results, and experimental metadata                                                              | Free        |
| Python Cryptographic Libraries      | Provides SHA-256, HMAC, data encoding, and integrity-related cryptographic operations                                                                                                | Free        |
| Pandas / CSV Processing Libraries   | Organises experimental records, controlled test data, result datasets, and evaluation outputs                                                                                        | Free        |

|                                                         |                                                                                          |                                                   |
|---------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|
| draw.io / diagrams.net                                  | Creates final research flowcharts, system architecture diagrams, and methodology figures | Free                                              |
| Microsoft Word / Institution-Provided Document Platform | Proposal and final research-report preparation                                           | Institution-provided / No additional project cost |

*Table 12 . Software and Development Platforms*

### 7.3 Human Effort Cost

The time and effort needed for system creation, controlled NFC tests, repeated verification trials, data analysis, documentation, and final evaluation constitute most the research's non-financial costs. Since the student researcher is performing this work as part of the academic project, the financial budget does not contain a separate labor expense.

### 7.4 Budget Justification

Because it allows for specific NFC observations and useful cross-reader verification without taking the place of the **Proxmark3** as the main research tool, the **Flipper Zero** is justified as a supplementary validation device. When reader-specific evidence is standardized, it is helpful to use a secondary device to see if identical NFC states result in consistent integrity interpretations.

The primary research tool needed for the suggested integrity-verification component is the **Proxmark3**. Supported NFC card technologies are identified, application/NDEF data is captured, readable raw memory is examined, supported security-related states are obtained, trusted baseline evidence is produced, and further verification scans are gathered. Thus, the experimental approach has a direct bearing on its functionality.

Because the suggested research is tested across **NTAG215**, **MIFARE Classic 1K**, and **MIFARE DESFire** rather than employing a single card technology, the NFC card set is necessary. The study may establish reliable baselines, carry out repeated, unaltered tests, implement controlled alterations, and assess family-specific tamper localization without the need for third-party credentials thanks to several authorized test cards.

Because the suggested prototype mostly uses Python, open-source NFC tooling, open-source data processing libraries, and free development tools, the software development cost is quite low. In addition to offering all software features needed for baseline protection, three-layer verification, tamper localization, journal validation, and experimental analysis, this keeps the financial cost concentrated on the actual NFC research equipment and test medium.

As a result, the research's total projected direct hardware cost is LKR 115,000, while the academic prototype's software and development platform costs are essentially negligible.

## 8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) divides the project into discrete functions for each team member while maintaining a common project timeframe. Each member is responsible for a distinct research component, including its design, implementation, testing, assessment, and documentation. Everyone collaborates to plan common tasks like setting up the lab, integrating the backend, creating dashboards, conducting final testing, and creating documentation. chronological congruence and equitable workload allocation.

| <b>Team Member</b> | <b>Main Research Component</b>                                      | <b>Main Responsibilities</b>                                                                                                                                                                                                                            | <b>Timeline Alignment</b>                                                                                 | <b>Workload Distribution</b>                                                                                   |
|--------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Member 1</b>    | Security Profile Scoring                                            | Analyse NFC security configurations, develop scoring logic, perform testing, evaluate results, and prepare component documentation                                                                                                                      | Development and testing performed in parallel with the other components                                   | Independent component development plus shared integration activities                                           |
| <b>Member 2</b>    | Physical / Timing Clone Detection                                   | Collect physical/timing evidence, develop clone-detection logic, perform controlled validation, and document results                                                                                                                                    | Uses the common laboratory and integration milestones                                                     | Independent research workload with shared hardware and integration tasks                                       |
| <b>Member 3</b>    | <b>NFC Tag Data Integrity Verification with Tamper Localization</b> | Develop trusted baseline creation, HMAC protection, application/NDEF verification, raw-memory verification, security-state verification, tamper localization, integrity journal, VALID / MODIFIED / UNKNOWN decision logic, and experimental evaluation | Developed in parallel with the other components and integrated during the common system-integration phase | Independent integrity-verification research plus shared backend, dashboard, and final testing responsibilities |
| <b>Member 4</b>    | Behavioural Anomaly Detection                                       | Collect access-event data, develop behavioural-analysis mechanism, perform validation, and document results                                                                                                                                             | Follows the same project milestones and final integration schedule                                        | Independent behavioural-analysis workload plus shared integration activities                                   |

|                    |                           |                                                                                                                                                                                            |                                                  |                                                                              |
|--------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------|
| <b>All Members</b> | Shared Project Activities | Literature review coordination, laboratory preparation, common backend/interface agreement, system integration, dashboard integration, testing, final report preparation, and presentation | Completed according to common project milestones | Shared tasks are distributed among members to maintain balanced contribution |
|--------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------|

Table 13 Work Breakdown Structure and Team Responsibility Distribution

## 9. GANTT CHART

The anticipated research timetable for the NFC Tag Data Integrity Verification with Tamper Localization project, which runs from June **2026 to June 2027**, is shown in the Gantt chart. The literature evaluation, research-gap analysis, finalization of the methodology, and laboratory preparation are the first steps in the timeframe. Data-acquisition, parsing, canonicalization, trusted-baseline, HMAC protection, three-layer verification, tamper-localization, and integrity-journal modules are then developed. Following the primary implementation tasks are planned controlled experimental testing, cross-reader validation, evaluation, and outcome analysis. While documentation is ongoing throughout the project, system integration and dashboard connection are planned for the latter stages. Final testing, revisions, dissertation completion, and presentation preparation are scheduled to end in June **2027**.

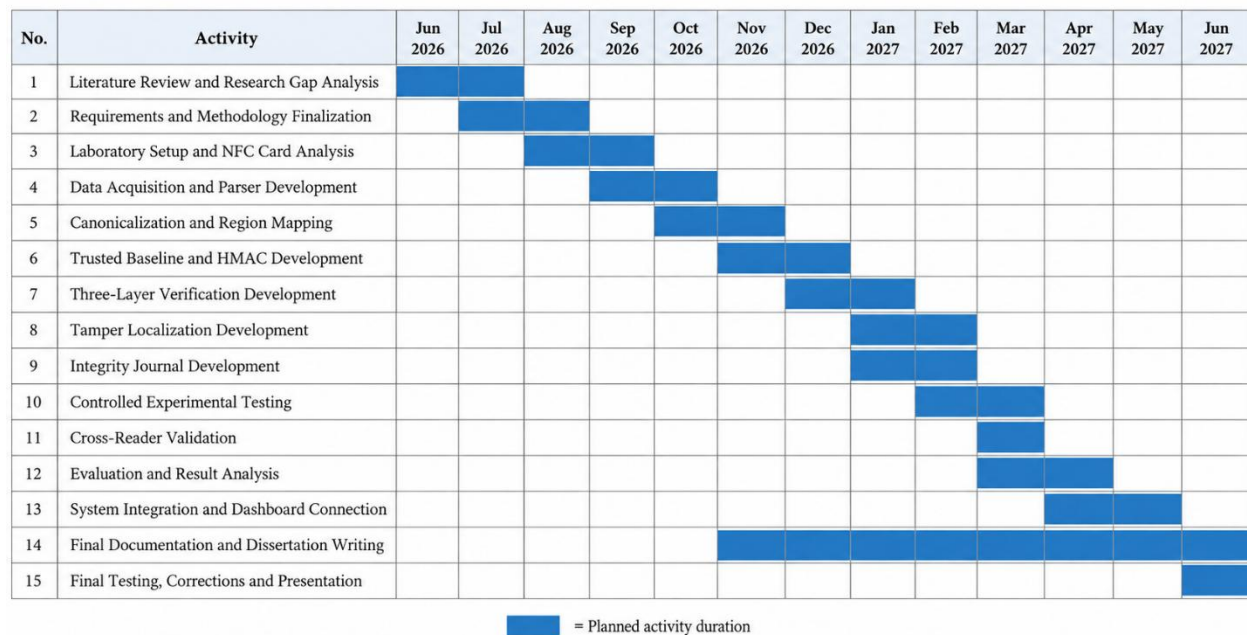


Figure 9 Project Gantt Chart

## 9.1 PROJECT TIMELINE

| Project Activity                                      | Month 1–2 | Month 3–4 | Month 5–6 | Month 7–8 | Month 9–10 | Month 11–12 |
|-------------------------------------------------------|-----------|-----------|-----------|-----------|------------|-------------|
| Literature Review and Problem Identification          | ✓         |           |           |           |            |             |
| Research Proposal Preparation                         | ✓         | ✓         |           |           |            |             |
| System Architecture Design                            |           | ✓         |           |           |            |             |
| Dataset Preparation and Card Analysis                 |           | ✓         | ✓         |           |            |             |
| Trusted Baseline and HMAC Design                      |           |           | ✓         |           |            |             |
| NFC Data Acquisition and Parser Development           |           |           | ✓         | ✓         |            |             |
| Verification Engine Development                       |           |           | ✓         | ✓         |            |             |
| Tamper Localization and Integrity Journal Development |           |           |           | ✓         | ✓          |             |
| Hardware Setup and Integration                        |           |           | ✓         | ✓         |            |             |
| Backend and Dashboard Development                     |           |           |           | ✓         | ✓          |             |
| System Integration                                    |           |           |           | ✓         | ✓          |             |
| System Testing and Performance Evaluation             |           |           |           |           | ✓          | ✓           |
| Result Analysis and Validation                        |           |           |           |           | ✓          | ✓           |
| Final Report Writing and Documentation                |           |           |           | ✓         | ✓          | ✓           |
| Final Corrections and Presentation Preparation        |           |           |           |           |            | ✓           |

Table 14 Project Timeline

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## 10. APPENDICES

Without interfering with the main chapters' flow, the appendices offer technical, experimental, ethical, and project-management evidence that supports the main idea. Detailed NFC card information, memory-region maps, experimental ground-truth records, baseline and journal structures, requirements evidence, laboratory procedures, sample verification outputs, interface designs, hardware evidence, project planning artifacts, and the faculty-mandated proposal evidence will all be included.

The appendices provide supporting technical, experimental, design, and project evidence related to the proposed **NFC Tag Data Integrity Verification with Tamper Localization** research.

### 10.1 Appendix 1 — Supporting Research Evidence

#### Appendix A — NFC Tag and Card Inventory

Contains the authorized **NTAG215**, **MIFARE Classic 1K**, and **MIFARE DESFire** cards used for controlled laboratory experiments.

#### Appendix B — NFC Card Family Memory Region Maps

Contains the relevant pages, sectors, blocks, applications, files, and security-related regions used during integrity verification.

#### **Appendix C — Controlled Modification and Ground-Truth Records**

Contains controlled unchanged and modified test cases, expected modification locations, and expected VALID, MODIFIED, or UNKNOWN results.

#### **Appendix D — Dataset and Trusted Baseline Structure**

Contains the structure used to store trusted baseline data, current scans, hashes, baseline versions, card-family information, and experimental metadata.

#### **Appendix E — Integrity Journal Structure**

Contains the proposed backend integrity-event journal and supported card-resident integrity evidence.

#### **Appendix F — Requirements Elicitation Evidence**

Contains actual laboratory observations, stakeholder feedback, supervisor/panel feedback, or other requirement-gathering evidence collected during the project.

#### **Appendix G — Experimental Test Procedure**

Contains the procedures used for enrolment, verification, controlled modification, degraded-read testing, and cross-reader validation.

#### **Appendix H — Sample Verification Reports**

Contains example VALID, MODIFIED, and UNKNOWN verification reports with localization evidence.

#### **Figure J.1. NFC Verification Dashboard with Tapping Count (Sample Interface)**

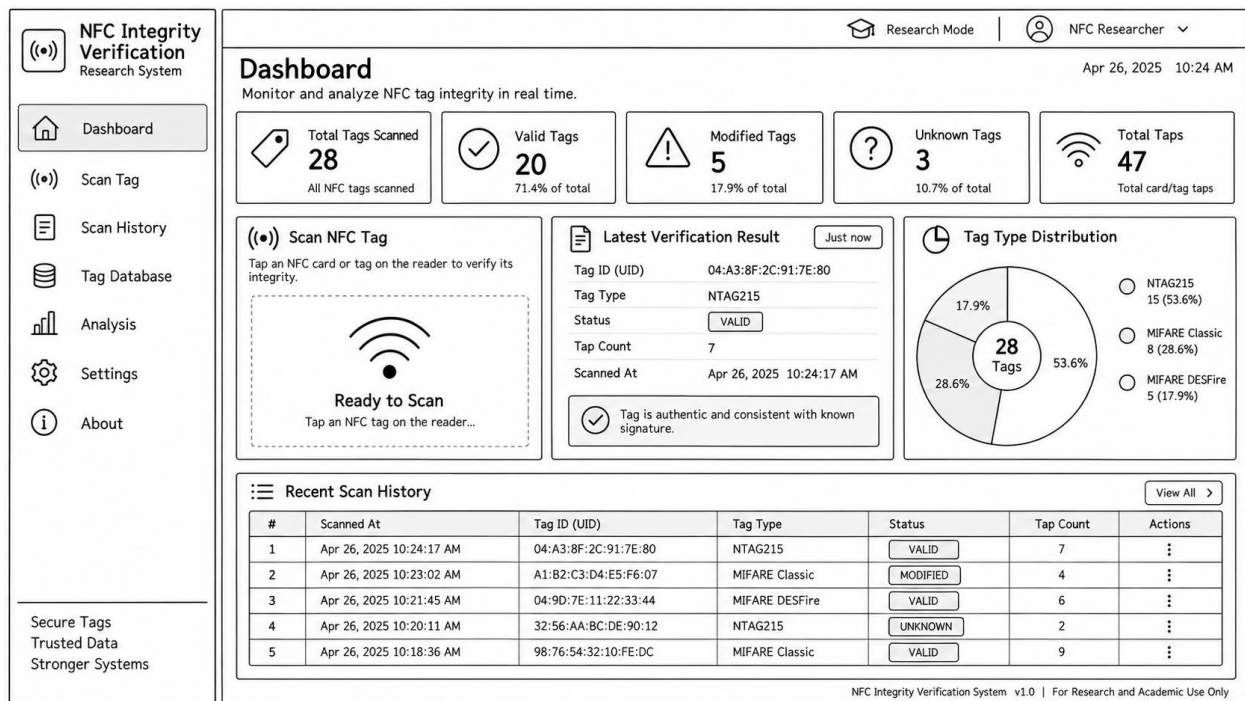


Figure 10 UI Dashboard

The dashboard displays:

- Total tags scanned
- Valid tags
- Modified tags
- Unknown tags
- **Observed tap count**
- Latest verification result
- Tag type
- Scan date and time
- Recent verification history

Important wording: use “**Observed Tap Count**” or “**System-Observed Tap Count**”, because the system can only count taps recorded by this verification system; it cannot prove the exact lifetime physical tap count of the NFC card.

## Appendix J — Hardware and Device Evidence

Contains supporting specifications for **Proxmark3**, **Flipper Zero**, and **NFC test cards/tags**.

## Appendix K — Ethics and Authorisation Evidence

Contains evidence that experiments are limited to owned or explicitly authorised NFC cards and tags.

## Appendix L — Work Breakdown Structure

Contains detailed WBS and individual responsibility allocation.

## Appendix M — Gantt Chart

Contains the complete project timeline from **June 2026 to June 2027**.

### 10.2 Appendix 2 — Mandatory Proposal Evidence and Decision Log

| Date               | Claim / Problem                                                                                                     | Evidence Collected                                                                   | Alternatives Considered                                                                   | Decision Made                                                                                                                         | Supervisor Discussion                                                                                        |
|--------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>26 Jun 2026</b> | Concern about importing and using Flipper Zero in Sri Lanka and whether regulatory approval is required             | Reviewed the device requirement and possible regulatory restrictions                 | Use Flipper Zero without confirmation / use another device / obtain regulatory permission | Confirm the relevant TRC/TRCSL requirements and obtain permission if required before importing or using the device                    | Supervisor advised checking with TRC/TRCSL and obtaining the necessary permission                            |
| <b>05 Jul 2026</b> | Whether Proxmark3 should be purchased for the research                                                              | Considering the NFC acquisition, scanning, and emulation requirements of the project | Use only Flipper Zero / use Proxmark3 as the primary research device                      | Use <b>Proxmark3 as the primary research device</b> because it supports detailed NFC scanning and research functions more effectively | Supervisor advised that Proxmark3 would make tapping, scanning, and emulation-related laboratory work easier |
| <b>22 Aug 2026</b> | Which hashing method is suitable for the integrity-verification component                                           | Compared hashing approaches for detecting changes in NFC data                        | SHA-256 and other hashing approaches                                                      | Use <b>SHA-256 as the main hashing method for initial integrity comparison and evaluation</b>                                         | Supervisor advised implementing SHA-256 first and evaluating its suitability experimentally                  |
| <b>13 Sep 2026</b> | A copied NFC card may contain the same data and therefore produce the same hash, so hashing alone cannot identify a | Discussed the limitation of hash-only verification and the need for                  | Hash-only comparison / additional persistent logging and history tracking                 | Add an <b>automatically generated persistent integrity log/journal</b> to record trusted events, updates,                             | Supervisor advised implementing an automatically running log mechanism that permanently stores               |

|  |                      |                              |  |                                      |                            |
|--|----------------------|------------------------------|--|--------------------------------------|----------------------------|
|  | byte-identical clone | persistent integrity history |  | timestamps, and verification history | relevant integrity records |
|--|----------------------|------------------------------|--|--------------------------------------|----------------------------|

### 10.3 Appendix 3 — AI Use Disclosure

| AI Tool        | Purpose of Use                             | What Was Generated / Assisted                                                                                                                                                                       | How Output Was Verified                                                                                                                                | Final Student Contribution                                                                                                                                                                    |
|----------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ChatGPT</b> | Research planning and proposal development | Assisted with structuring the research problem, research gap, objectives, methodology, evaluation strategy, user requirements, and proposal sections.                                               | Suggested content was reviewed against research papers, project scope, supervisor feedback, and proposal requirements.                                 | The student selected, modified, and finalized the research content according to the actual project and supervisor guidance.                                                                   |
| <b>ChatGPT</b> | Academic writing and language improvement  | Assisted in improving grammar, clarity, academic wording, paragraph structure, and organization of proposal content.                                                                                | The revised text was reviewed by the student to ensure that the meaning, research scope, and technical information remained accurate.                  | The student made the final decisions regarding wording, technical descriptions, research claims, and document content.                                                                        |
| <b>ChatGPT</b> | Technical and system-design support        | Assisted in explaining trusted baseline creation, SHA-256, HMAC, three-layer verification, tamper localization, integrity journaling, VALID / MODIFIED / UNKNOWN decisions, and evaluation metrics. | Technical suggestions were checked against selected research literature, NFC card specifications, planned experiments, and supervisor recommendations. | The student determined the final system architecture, technologies, implementation scope, experimental design, and research methodology and is responsible for implementation and evaluation. |